

Question	Marking guidance	Mark	Comments
03.1	At least 4 straight vertical lines with even spacing ✓ Arrows pointing upwards ✓	2	Condone reasonable freehand drawing
03.2	Use of $E = \frac{V}{d}$ ✓ $6.0 \times 10^5 \text{ (N C}^{-1}\text{)}$ ✓	2	Condone POT error for first mark
03.3	2 marks for use of 3 formulae; 1 mark for use of 2 formulae: $C = \frac{A\epsilon_0\epsilon_r}{d}$ $C = \frac{Q}{V}$ $E = \frac{V}{d}$	2	Must see a final expression of $\frac{Q}{\epsilon_0\epsilon_r A}$ Zero marks for use of radial field formula.
03.4	Attempt to determine an area using dimensions in Fig. 5 ✓ $3.2 \times 10^{-7} \text{ (C)}$ ✓	2	Allow ecf for their 03.2

Question	Marking guidance	Mark	Comments
03.5	Insulating ✓ Polarizable ✓	2	Allow descriptions of "polarizable"
03.6	E reduced ✓ so pd decreases as d constant ✓ OR C increased ✓ so pd decreases as Q constant ✓	2	

Question	Marking guidance	Mark	Comments
----------	------------------	------	----------

9	C
---	---

18	C
----	---

	21	B
--	----	---

	22	A
--	----	---

	23	A
--	----	---

	32	A
--	----	---

	33	C
--	----	---

Question	Marking guidance	Additional comments/guidelines	Mark
02.1	Reads off graph (-7.4) and takes exponential to get Q_0 6.1×10^{-4} (C) ✓		2
02.2	Uses $C = \frac{Q}{V}$ to produce 1.22×10^{-4} (F) ✓		1
02.3	Extracts data correctly from the graph ✓ Correct use of data ✓ 79 (s) ✓	Methods may use gradient (1.27×10^{-2}), $T_{1/2}$ (55 s), time when $Q=0.37Q_0$, or a pair of readings from the graph and substitution into $Q = Q_0 e^{-\frac{t}{RC}}$. Gradient method must use $t \geq 150$ s.	3
02.4	Uses time constant = RC (expect $0.66 \text{ M}\Omega$) ✓ Uses parallel resistor formula ✓ $0.84 \text{ (M}\Omega\text{)}$ ✓	Allow any answer that rounds to $0.8 \text{ (M}\Omega\text{)}$ from different intermediate values for C or time constant.	3
02.5	R is greater (in Fig 4) ✓ States time constant is larger, or calculates larger time constant OR References $V = IR$ with same V ✓ Lower (initial) rate of discharge or smaller (initial) current, from some reasoning ✓	Allow MP1 and MP2 for correct calculations of both initial currents. (Fig 3: $7 \mu\text{A}$; Fig 4: $2 \mu\text{A}$)	3
Total			12

Question	Marking guidance	Additional comments/guidelines	Mark
----------	------------------	--------------------------------	------

Question	Marking guidance	Additional comments/guidelines	Mark
03.1	1.1×10^4 (N) ✓		1
03.2	Vector addition of L and W ✓ Horizontal arrow drawn pointing right ✓	Allow parallelogram. W must be vertical.	2
03.3	Vertical component of lift = weight OR $L \cos 25$ (= W) seen ✓ 2 sf value that rounds to 1.2×10^3 (N) ✓	Allow $L \sin 65$ or $L \cos \theta$ ecf from 03.1	2
03.4	Horizontal component of lift = centripetal force or $L \sin \theta = F_c$ OR uses $R^2 = L^2 - W^2$ ✓ Evaluates F_c (5070 N) OR uses centripetal force formula ✓ Allow range of 63 to 66 (m s^{-1}) ✓	Condone use of L in centripetal force equation for MP2 Allow 2 marks for 99 (m s^{-1}) and ecf for their 03.3.	3
03.5	(Aircraft) decreases height/altitude or gains vertical speed downwards or accelerates downwards ✓ (Because) vertical component of lift < weight ✓ (Aircraft) moves in smaller (arc) radius ✓ (Because horizontal component of lift increases, so) centripetal force increases ✓	Don't credit "L increases" Don't credit "L increases". Allow reference to centripetal force equation.	4
Total			12

Question	Marking guidance	Additional comments/guidelines	Mark
----------	------------------	--------------------------------	------

11	C
----	---

16	C
----	---

	21	B
--	----	---

	22	A
--	----	---

	23	B
--	----	---

	24	C
--	----	---

	25	C
--	----	---

	33	D
--	----	---

Question	Answers	Additional comments/Guidelines	Mark	AO
06.1	use of $E = \frac{1}{2} CV^2$ leading to 6.1×10^{-2} (J) ✓	at least 2sf	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
06.2	Use of $V = V_0 e^{-\frac{t}{RC}}$ ✓ 1.5 (V) ✓		2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.3	(half the) energy is dissipated in the resistor (in the charging circuit) ✓ reference to equation $E = \frac{1}{2}CV^2$ OR reference to area under QV graph being $\frac{1}{2}QV$ ✓	condone in the internal resistance of the battery for mp1	2	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
06.4	converts 50 mW h into J correctly (= 180 J) ✓ halves (= 90 J) and divides by 0.06 J to give 1500 ✓	unrounded version gives 1480 to 3 sf ecf from 06.1	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.5	Max 3 from greater duration OR brightness decreases more gradually ✓ because greater time constant (= RC) ✓ (initial) brightness is the same ✓ because initial voltage OR initial current is the same ✓	Reason must be related to the correct effect	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.6	fewer flashes can be delivered ✓ because the energy stored in the capacitor is greater ✓	MP2 is dependent on MP1	2	AO2
Total			12	

Question	Key
----------	-----

8	C
---	---

9	B
---	---

11	D
----	---

15	D
----	---

16	C
----	---

17	D
----	---

18	A
----	---

Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	Correct substitution of Q and r into $F = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{r^2} \checkmark$ To give $5.8 \times 10^{-9} \text{ (N)} \checkmark$	Allow POT errors in substitution Correct answer gets both marks.	2	1 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	Evidence of use of potential at a point $V = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$ Or potential energy equation $\checkmark$ Attempt to calculate potential energy difference = (their $\Delta V \times Q$) $\checkmark$ $1.2 \times 10^{-10} \text{ (J)} \checkmark$	For MP1 expect to see calculation of - Potential at 1.5 mm for one of the charges (7.2 V) - Or - Potential energy of charges at 1.5 mm separation ($8.6 \times 10^{-12} \text{ J}$) - Or - Potential at $101 \times 10^{-6} \text{ m}$ for one of the charges (107 V) - Or - Potential energy of charges at $101 \times 10^{-6} \text{ m}$ separation ($1.3 \times 10^{-10} \text{ J}$) Potential difference (ΔV) = 100 V Condone POT errors in MP1 and MP2 Do not credit use of work done = force $\times$ distance	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
04.3	(vertically) down ✓		1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
04.4	Evidence of calculation of weight of droplet ✓ Use of $F = EQ$ ✓ To give $E = 6.4(5) \times 10^3 \text{ (V m}^{-1}\text{)} \checkmark$	Expect to see $F = mg = 1.38 \times 10^{-9} \times 9.81 = 1.35 \times 10^{-8} \text{ (N)}$ MP2 can be given for substitution or rearrangement. Allow ecf for their F Condone minus sign in final answer	3	1 × AO1 2 × AO2

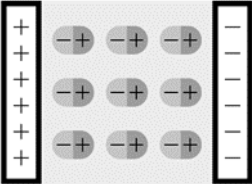
Question	Answers	Additional comments/Guidelines	Mark	AO
04.5	Calculation of area ✓ evidence of $C = \frac{A\epsilon_0\epsilon_r}{d}$ to give $1.55 \times 10^{-7} \text{ F} \checkmark$	Expect to see $5.9 \times 10^6 \text{ m}^2$ At least 3 sf	2	1 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
04.6	Use of energy stored = $\frac{Q^2}{2C}$ ✓ 2.4×10^{10} (J) ✓	Allow ecf from 4.5 Show that value gives 2.5×10^{10} (J)	2	AO2
Total			13	

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----

10	A	circular, curving upwards	AO2
----	---	---------------------------	-----

12	D	- 86%	AO2
----	---	----------	-----

15	C		increases	AO1
----	---	--	-----------	-----

Question	Marking guidance	Mark	Comments
05.1	37 milli coulombs (stored) per volt ✓	1	Allow 37mC and 1V instead of full words

Question	Marking guidance	Mark	Comments
05.2	The capacitance doubles and reference to $C = \frac{\epsilon A}{d}$ ✓ voltage stays the same ✓ statement of $Q = CV$ and Q doubles ✓	3	

Question	Marking guidance	Mark	Comments
05.3	Idea that an equal number of electrons is removed from the positive plate as is deposited onto the negative plate ✓	1	Condone idea that electrons transferred from +ve plate to -ve plate (through cell)

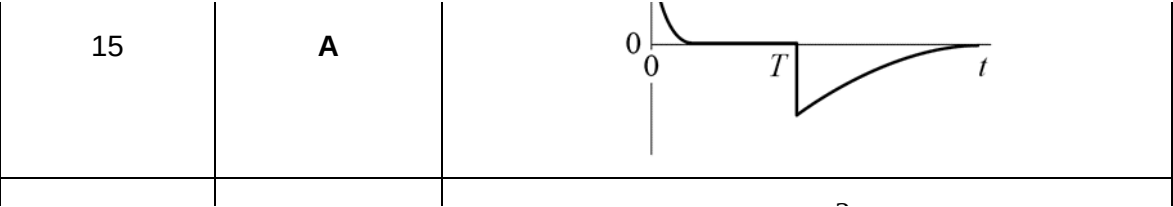
Question	Marking guidance	Mark	Comments
05.4	Q stays the same ✓ as the capacitor is isolated OR charge cannot flow ✓ The capacitance halves AND reference to $E = \frac{Q^2}{2C}$ leading to energy doubles ✓	3	Note: if another version of the energy equation is used, eg $E = \frac{1}{2}CV^2$, only award full marks if the candidate states that V doubles MP3 is dependent on MP1 Allow 2 marks for a reasoned qualitative argument not justified by equations, ie work has to be done by external system to separate the + and – plates. ✓ Some energy must be stored in the capacitor so the energy stored increases. ✓

Question	Marking guidance	Mark	Comments
05.5	Idea that (energy is conserved because) the gain in stored energy is provided by the work done in pulling apart the plates ✓	1	
Total		9	

Question	Marking guidance	Mark	Comments
----------	------------------	------	----------

11	C	$3\pi^2GR^4\rho^2$
----	----------	--------------------

13	B	$\frac{4\pi\epsilon_0 G A m_e M_n}{Ze^2}$
----	----------	---



16	B	$RC \ln \frac{3}{2}$
----	----------	----------------------

19	A	$Bv = E$
----	----------	----------

Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	Evidence of use of $R = V/I$ $= 1.6 \times 10^4 \Omega$ ✓	$R = 6/3.8 \times 10^{-4}$ Calculator gives 1.578947×10^4	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	Idea that the pd across capacitor increases therefore pd across variable resistor must decrease ✓ (So R must decrease to maintain constant I)		1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
04.3	$Q = It$ or equivalent seen To give $Q = 3.8 \times 10^{-4} \times 120$ $= 0.0456 \text{ (C)}$ ✓	Do not accept 2 sf answer Working must be seen	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
04.4	Reading from graph to obtain V after 120 s AND Use of $C = \frac{Q}{V}$ = their answer to 04.3 divided by their V ✓ To give 8.1×10^{-3} F ✓	Accept 5.5 V to 5.7 V The second mark is a quality mark. Accept 8.0 to 8.3 Allow use of 0.046 C to give 8.1 to 8.4×10^{-3} F Alternative using $C = \frac{\text{current}}{\text{gradient}}$. MP1 for determination of gradient using at least half of the line. MP2 for final answer in range.	2	1 × AO1 1 × AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
04.5	Use of $C = \frac{\epsilon_0 \epsilon_r A}{d}$ ✓ (To give $\epsilon_r = 0.13 \times 10^{-9} \times 3.5 \times 10^{-3} / (8.85 \times 10^{-12} \times 0.15^2)$) = 2.3 ✓	Either for substitution or manipulation Condone PoT in MP1	2	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.6	Idea that the capacitance is too small/ much lower (than original)✓ No because: Charge stored too small/much less (at the same pd) ✓ Current would become zero too quickly to take readings (with a stop watch) OR pd across capacitor would reach 6.0 V too quickly to take readings (with a stop watch) ✓	Alternative: For MP1 makes use of capacitance in a valid equation without PoT error✓ For MP2 determines the charge stored✓ For MP3 determines the time and makes suitable comment. ✓ Alternatives for MP3 idea that pd changes too quickly for the current to be kept constant by adjusting the variable resistor. Alternative for MAX 2 Yes but: Would need much larger resistance ✓ Must have a much more sensitive ammeter ✓	3	AO4
Total			10	

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----

13	A	increases stays the same	AO1
----	----------	-------------------------------	-----

14	D	current in R potential difference across P	AO3
----	---	--	-----

15	B	15 s	AO2
----	----------	------	-----

16	c	120 mJ	AO1
----	----------	--------	-----

Question	Answers	Additional comments/Guidelines	Mark	AO
02.1	(Magnitude of the resultant) force (acting on the body) is proportional to its displacement (from its equilibrium position) ✓ and is always directed towards the equilibrium position ✓	Accept acceleration for force. Accept 'in opposite direction to displacement'	2	2 AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
02.2	Out of the page ✓ Reference to Fleming's left-hand rule ✓	Condone 'to the right' if the candidate is clearly referring to Figure 6 Do not condone "upwards" for "out of page." Accept "FLHR" or "left hand rule"	2	1 AO1 1 AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
02.3	Use of $F = ma$ or $l = 2\pi rN$ ✓ Use of $F = BIL$ ✓ 0.81 (T) ✓	evidence for MP2 can be seen as values 4.2(N) or 3.0(m) by substitution or manipulation	3	1 AO1 2 AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.4	Use of $a_{\max} = \omega^2 A$ ✓ $2.0(3) \times 10^{-4}$ (m) ✓	Ignore any minus sign	2	1 AO1 1 AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.5	(Use of $v = \omega A$ leading to) 0.33 (m s ⁻¹) ecf from 02.4 ✓		1	1 AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
02.6	Minus cos graph shown for $1\frac{1}{2}$ cycles ✓ 3.8×10^{-3} (s) shown after 1 cycle and candidate's value from 02.5 shown on ordinate ✓		2	2 AO3
Total			12	

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----

Question	Answers	Additional comments/Guidelines	Mark	AO
01.1	Arrow(s) upwards with suitable label e.g. reaction or N or R AND Single arrow downwards with suitable label e.g. weight or W or mg ✓ Starting point of downward arrow near to centre (of mass) of person AND Starting point of two upwards forces at feet of person AND Length of downward arrow about double that of upward arrows ✓	Condone alternative labels e.g. 'normal', 'contact force', 'gravity', 'gravitational (force)'. Reject 'F' MAX 1 if there are any horizontal forces e.g. friction. Reject answer that includes extra vertical forces. The judgement about the length is by eye.	2	AO1 AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
01.2	Arrow, the line of action of which is directed through the person towards centre of station ✓	Ignore labels Reject arrows that are too short or wobbly to prevent judgement of direction.	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
01.3	Idea that people (on Earth) experience weight due to reaction force (from “ground”) ✓ Links reaction force to centripetal force/acceleration.	Allow MP1 for answers that refer to simulation of gravity and refer to the reaction (from spacestation wall). Allow ‘normal force’ for ‘reaction force’ Allow ‘force towards the centre’ for centripetal force.	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
01.4	Equates g to an equation for centripetal acceleration ✓ Evidence of v OR ω , OR use of their v in $p = mv$ or ω in $p = mrv\omega$ ✓ 65 (kg) ✓	Can be seen by substitution or symbols. Look for $v = 38 \text{ m s}^{-1}$ Allow use of 148 to 150 m for ‘ r ’ Expect answer that rounds to 65 (kg) unless 148 used to give an answer rounding to 66 (kg)	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
01.5	Idea that ball will move in a straight line to the right/ maintain a constant velocity/ maintain its momentum ✓ idea that there are no forces acting on ball ✓ Explanation in terms of Newton 1 or 2 ✓	Reward evidence seen on diagram. Allow MP2 for idea that there is no centripetal force if supported by MP1 or MP3.	3	AO2
Total			11	

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----

Question	Answers	Additional comments/Guidelines	Mark	AO
05.1	Idea that the shaded area is the charge moved (in 6 s) ✓ Charge remaining = initial charge – charge moved ✓	Condone 'total' for 'initial' Do not accept 'total area' for initial charge.	2	AO3 AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
05.2	Reads off current at $t = 0$ OR <i>Their answer for $emf = IR$ for their max I</i> ✓ 6.3 V ✓	Allow 28 to 29 (μA) Accept POT error in MP1 Do not allow answer based on area under the curve. Accept 6.2 to 6.4 V	2	AO3 $\times$ 1 AO2 $\times$ 1

Question	Answers	Additional comments/Guidelines	Mark	AO
05.3	Attempts to determine RC from the graph OR evidence of use of discharge equation OR use of 'half-life' ✓ 1.2×10^{-5} (F) ✓	Expect to see an RC that rounds to 3 s ✓ Allow 1.0 to 1.4×10^{-5} consistent with their read-offs. If no other mark awarded, allow MAX 1 for evidence of estimation of area under graph leading to an answer of 1.0 to 1.4×10^{-5} (F)	2	AO3 $\times$ 1 AO2 $\times$ 1

Question	Answers	Additional comments/Guidelines	Mark	AO
05.4	Complete circuit with either ammeter or resistor' in series with capacitor connected to moving part of switch ✓ Complete circuit with both ammeter and resistor in series with capacitor on moving part of switch ✓		2	AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
05.5	Line with decreasing negative gradient ✓ that follows the dotted line on the graph ✓	MP2 is contingent on MP1 Do not condone line that ends abruptly at the time axis. In MP2 do not condone line that is consistently above or consistently below the dashed line.	2	AO2
Total			10	

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----

10	D	2.32F	AO2

12	B	there is no force	there is a force	AO1
----	---	-------------------	------------------	-----

14	C	2.5	0.9	AO1

15	C	circular	straight	AO1
----	---	----------	----------	-----

17	B	Q	$< E$	AO2
----	---	-----	-------	-----

Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	(use of intercept to give) $6.2 \pm 0.05 \text{ (V)}$ ✓		1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	Evidence of attempt to use $R = V \div I$ ✓ $6.5 \times 10^3 \text{ (}\Omega\text{)}$ ✓	For mp1 expect to see $-1 \times \text{gradient}$ or $\frac{\varepsilon - V_c}{I}$ for one point or $\varepsilon \div I$ -intercept For MP2 accept 6.3×10^3 to 6.7×10^3	2	1 × AO1 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.3	Idea that smaller current means less charge moved per second (and therefore smaller increase in V_c per second)	Accept use of charging equation and idea that rate of change of voltage decreases.	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO																						
04.4	Uses pair of current points OR emf and V point✓ Uses appropriate time between points ✓ Substitutes into charging equation for current or voltage ✓ (time constant =) 1.4×10^{-2} (s) based on correct working ✓ Use of time constant = their answer to 04.2 × C ($C = 2.2 \times 10^{-6}$ (F)) ✓	In MP1 condone use of any two V points Allow half-life approach NEW MS Allow 1.3×10^{-2} to 1.5×10^{-2} for MP4 In MP3 V_o must be the emf: $V = V_o(1 - e^{-\frac{t}{RC}})$ OR $I = I_o e^{-\frac{t}{RC}}$ <table><tr><th>I/mA</th><th>V/V</th></tr><tr><td>0.95</td><td>0.00</td></tr><tr><td>0.67</td><td>1.83</td></tr><tr><td>0.47</td><td>3.12</td></tr><tr><td>0.33</td><td>4.03</td></tr><tr><td>0.24</td><td>4.67</td></tr><tr><td>0.17</td><td>5.12</td></tr><tr><td>0.12</td><td>5.46</td></tr><tr><td>0.08</td><td>5.66</td></tr><tr><td>0.06</td><td>5.82</td></tr><tr><td>0.04</td><td>5.93</td></tr></table> Credit 1/e or ‘half-life’ approach for MP1, MP2, MP3 e.g. Uses .6.2 V and 3.1 V✓ Uses $T_{1/2} = 2 \times 5 \text{ ms}$ ✓ Uses time constant = $\frac{T_{1/2}}{\ln 2}$ ✓	I/mA	V/V	0.95	0.00	0.67	1.83	0.47	3.12	0.33	4.03	0.24	4.67	0.17	5.12	0.12	5.46	0.08	5.66	0.06	5.82	0.04	5.93	5	3 × AO3 2 × AO2
I/mA	V/V																									
0.95	0.00																									
0.67	1.83																									
0.47	3.12																									
0.33	4.03																									
0.24	4.67																									
0.17	5.12																									
0.12	5.46																									
0.08	5.66																									
0.06	5.82																									
0.04	5.93																									

Question	Answers	Additional comments/Guidelines	Mark	AO
04.5	Straight line from origin ✓ Ending at (0.95, 6.2) ✓	Allow half a grid square Allow ecf from their Figure 7 . If line not drawn in Figure 7 allow ecf from 04.1	2	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.6	Idea that the time interval (5 ms) is too short for the ammeter and voltmeter readings to be recorded manually ✓		1	AO4

Total			12	
--------------	--	--	-----------	--

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----

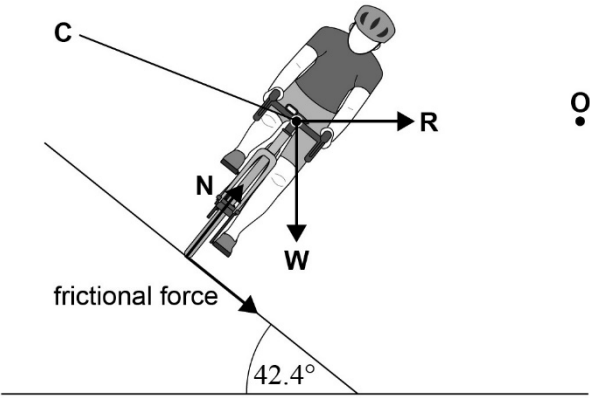
Question	Answers	Additional comments/Guidelines	Mark	AO
05.1	Two from ✓✓ • determines decay constant = $4.18 \times 10^{-9} \text{ s}^{-1}$ • $N = 62 \text{ (kBq)} \div \text{their decay constant}$ • mass = $60 \times \text{their } N \div 6.02 \times 10^{23}$ $1.48 \times 10^{-9} \text{ (g)} \checkmark$	Allow 2 sf answer	3	1 × AO3 2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.2	Attempts to use $A = A_0 e^{-\text{their } \lambda t} \checkmark$ Correct use of $\log_e \checkmark$ 15(years) ✓	Allow ecf for their λ	3	1 × AO1 2 × AO2

Total			6	
-------	--	--	---	--

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----

Question	Answers	Additional comments/Guidelines	Mark	AO
03.1	Maximum frictional force = $\frac{mv^2}{r} = \frac{79.5 \times 14.2^2}{22.5} = 712 \text{ N} \checkmark$	At least 3 sf required	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
03.2	Any two from: $\checkmark$ - normal acting at right angles to the surface where the tyre meets the road - weight acting downwards from centre of mass - resultant (centripetal) force acting horizontally to the right, touching rider or bike All three $\checkmark$		2	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
03.3	Finds horizontal component of $N \checkmark$ Resolves towards O and attempts to set up equation $N \sin 42.4^\circ + F_2 = \frac{mv^2}{r} \checkmark$ Shows that $F_2 = 11.2 \text{ (N)}$ and $11.2 < 710 \checkmark$	Allow ecf for wrong trig function in mp2 Allow use of $F_2 \cos 42.4^\circ$	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
03.4	Component of normal contact force contributes to the centripetal force ✓ As the speed increases the required centripetal force increases ✓ The maximum speed on the flat is when max friction = centripetal force ✓ The maximum speed on the bank is when (horizontal component) max friction + (horizontal) component of reaction force = centripetal force which can be greater than maximum frictional force ✓	Condone 'provides' for 'contributes to'	4	AO2
Total			10	

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----

Question	Answers	Additional comments/Guidelines	Mark	AO
06.1	$(1 - e^{-1}) 0.63$ ✓	Allow 63%	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
06.2	Reading of 5.5 (mA) for y -intercept and use of $I = \frac{V}{R}$ ✓ 1100 (Ω) to at least 2 sf ✓		2	1 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
06.3	Use of $I = I_0 e^{-\frac{t}{RC}}$ with data from graph ✓ Time constant = 8.0–8.4 (s) ✓ Capacitance = $\left(\frac{\text{their time constant}}{\text{their answer to 06.2}} \right)$ ✓		3	AO3

Question	Answers		Additional comments/Guidelines	Mark	AO
06.4	Mark	Criteria	Changes from $t = 0$ to $t = 7$ s - capacitor charges - voltage increases - larger charge on capacitor Relationship between meter readings - voltage across capacitor increases so voltage across resistor decreases, so current falls - making it more difficult for charge to be added to capacitor, so current falls - current = voltage across resistor / resistance of resistor - emf = voltage across capacitor (voltmeter reading) + voltage across resistor Changes after $t = 7$ s - current flows in opposite direction when discharging hence negative - capacitor discharges - voltage across capacitor falls so voltage across resistor falls, so current falls - charge falls on capacitor - so less force on pushing charges round circuit so current falls	6	2 × AO2 4 × AO3
	6	All three areas covered in some detail. 6 marks can be awarded even if there is an error and/or parts of one aspect missing. However, for 6 marks an explanation for the negative current after 7 s must be present.			
	5	All three areas covered, at least two in detail. Whilst there will be gaps, there should only be an occasional error.			
	4	Two areas successfully discussed, or one discussed and two others covered partially. Whilst there will be gaps, there should only be an occasional error.			
	3	One area discussed and one discussed partially, or all three covered partially. There are likely to be several errors and omissions in the discussion.			
	2	Only one area discussed or makes a partial attempt at two areas.			
	1	None of the three areas covered without significant error.			
	0	No relevant analysis.			
Total				12	

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----

12	A	A s V^{-1}	AO1
----	---	---------------------	-----

13	C	0.08 3.6	AO2
----	---	----------	-----

Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	3.7 (J) ✓	$E = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} \times \frac{(3.7 \times 10^{-2})^2}{185 \times 10^{-6}} = 3.7 \text{ J}$	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	Use of $C = \frac{Q}{V}$ or $V = IR$ ✓ 7.4 (A) ✓	Use of another capacitance equation to work out V is also appropriate for mp1, eg $E = \frac{1}{2} CV^2$ $V = \frac{Q}{C} = \frac{3.7 \times 10^{-2}}{185 \times 10^{-6}} = 200 \text{ V}$ $I = \frac{V}{R} = \frac{200}{27} = 7.4 \text{ A}$	2	AO2

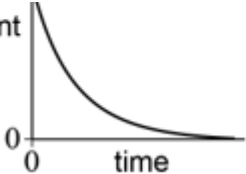
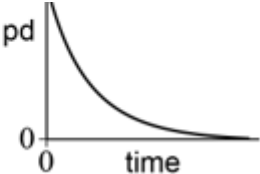
Question	Answers	Additional comments/Guidelines	Mark	AO
04.3	Any four from: ✓✓✓✓ (explanation of initial rise) - the idea that current causes an increase in temperature of the lamp - the idea that the temperature rise is due to collisions between electrons and ions / atoms - the idea that the initial rise in intensity is due to the initial rise in temperature (explanation of later decrease) - the idea that the current from the discharging capacitor decreases with time - the idea that the decrease in temperature decreases the intensity of the light emitted.	condone power for current in bp4	4	2 × AO2 2 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.4	At least three read-offs from graph used (to find two ratios or test half-life twice) and all read off correctly to within half a square ✓ Valid test for exponential decrease and valid conclusion that it is not exponential decrease ✓	Example 1 – ratio test at two fixed intervals of time: After 4 divisions of time from peak, intensity changes from 10 divisions to 5 After 4 further divisions from peak, intensity changes from 5 divisions to 4 divisions $\frac{10}{5} = 2 ; \frac{5}{4} = 1.25$ Values too different and so exponential decrease is not occurring Example 2 – testing time for half-life After 4 divisions of time from peak the intensity halves from the peak of 10 divisions to 5 divisions After a further 4 divisions of time from the intensity of 5 divisions the intensity has only changed to 4 divisions so has not halved again Values for half-life too different and so exponential decrease is not occurring	2	1 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.5	Resistance decreases which will make the discharge take less time, so the flash will not last as long / student is incorrect ✓	Allow “will make time constant shorter” or “will make the discharge quicker”	1	AO2
Total			10	

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----

14	D	14 s	AO2
			AO3

15	A	current 0 time  pd 0 time 	
----	---	--	--

16	A	$\frac{1}{2} C \varepsilon^2$	AO1
----	---	-------------------------------	-----

Unknown Unknown - Unit 3 - Q02

question	marking guidance	mark	comments
02.1	$(3.0 \times 10^{-10}/24) \times 6.02 \times 10^{23}$ seen ✓ $7.52(5) \times 10^{10}$ Answer seen to at least 3 sf	1	
02.2	Decay constant = $(0.69 / 14.8 \text{ h}^{-1})$ or $1.3 \times 10^{-5} \text{ s}^{-1}$ ✓ $A = 1.30 \times 10^{-5} \times 7.5 \times 10^{10}$ ✓ $9.75 \times 10^5 \text{ (Bq)}$ ✓ Allow 2 or 3 sf Allow use of $A = \lambda N$ with an incorrectly calculated decay constant	3	
02.3	Activity 3.5 h later should be $A = 9.8 \times 10^5 \text{ e}^{-0.0466 \times 3.5}$ ✓ $8.33 \times 10^5 \text{ (Bq)}$ ✓ Volume of liquid = $(8.33 \times 10^5 / 3600) \times 15 = 3470 \text{ (cm}^3\text{)}$ ✓	3	
02.4	Estimate gives 3700 compared with 3500 ✓ Flask has more mass than average / liquid is not water ✓	2	

question	marking guidance	mark	comments
----------	------------------	------	----------

Unknown Unknown - Unit 3 - Q07

question	marking guidance	mark	comments
7.1	$28 \times 10^{-6} \text{ [C]}$ ✓	1	
7.2	Attempted use of $\frac{1}{2}CV^2$ or $\frac{1}{2}QV$ and readoff at 50 s correct ($1.2 \pm 0.1 \text{ V}$) ✓ $3.4 \times 10^{-6} \text{ [J]}$ ✓ Difference between 85 μJ and candidate energy ✓	3	
7.3	Time constant is time taken to fall to $\frac{V_0}{e} = \frac{6.0}{2.72} = 2.21 \text{ V}$ ✓ reads off time constant as 32 s ✓ Time constant is RC ✓ $R = 6.8 \text{ M}\Omega$ ✓ OR Use of $V = V_0 e^{-\frac{t}{RC}}$ ✓ V readoff correct for a $t > 25 \text{ s}$ and correct substitution ✓ $RC = 32 \text{ s}$ ✓ $R = 6.8 \text{ M}\Omega$ ✓	4	

this section, each correct answer is awarded 1 mark.

Question	Key
----------	-----

Question	Key
----------	-----